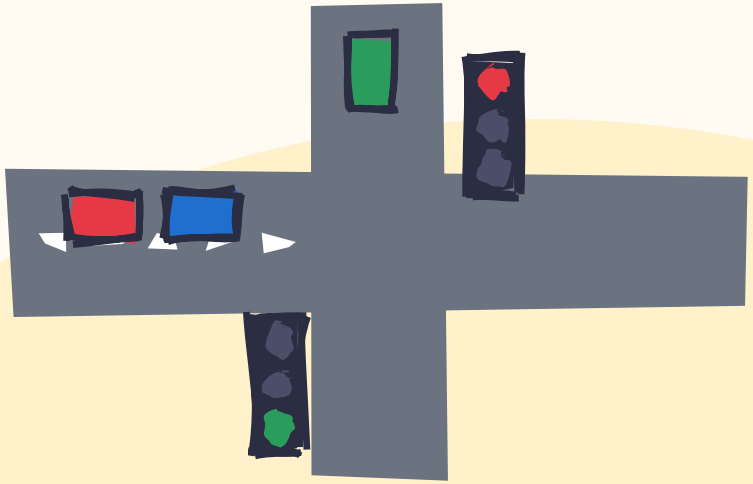
Junior Makers · Class 25

Smart Traffic Junction with Car Counter

Lights that count cars and think for themselves

Safe sequence | IR car counter | Smart decisions | OLED display



Did you know?
Some cities use smart traffic lights that cut waiting times by about a quarter, saving fuel and reducing pollution.

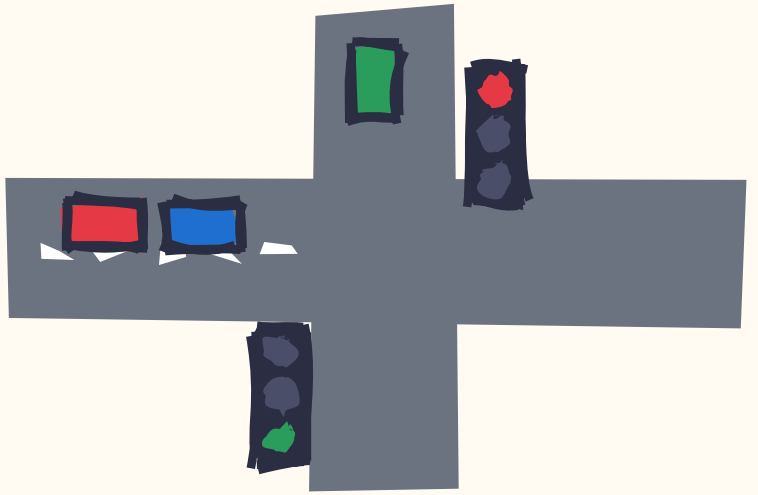
LEARN

Let's think...

Why wait when the other road is empty?

Annoying!

You're stuck at a red light. The other road has **no cars at all**, but it still gets a long green. Ordinary lights just use a **timer**. What if the lights could **see** the cars and decide who goes first?



Longer greens?

A police officer?

Smart sensors!

Today: you build a cardboard junction with **two traffic lights**, **car counters**, and code that gives green to the busier road.

Learning objectives

By the end of this class I can...

1	**Safe sequence:** program a traffic light **sequence** where both roads are **never green together**	**Proof:** my two lights change safely, every time
2	**Count cars:** count cars with an **IR sensor**, adding 1 only when a car **arrives**	**Proof:** pushing 5 toy cars past gives a count of 5
3	**Smart decision:** give green to the **busier road** while keeping it **fair**	**Proof:** the road with more cars gets green first, but nobody waits forever
4	**Test & show:** show counts on an **OLED** and test the junction with traffic	**Proof:** our test table for 3 traffic situations

Objective 1 · Safe sequence · video

Video time!



How do traffic lights work?

Fun Kids Learn · about 4 minutes · youtu.be/nWY7ZcOjGXc

Watch for...

1. How do some traffic lights **know** a car is waiting?
2. Why is there an **amber** light?
3. Who controls the traffic lights in a city?

Scan to watch

After the video: Think of a junction near school. Is it on a timer, or does it sense cars?

Objective 1 · Safe sequence

The golden rule: never both green

The idea

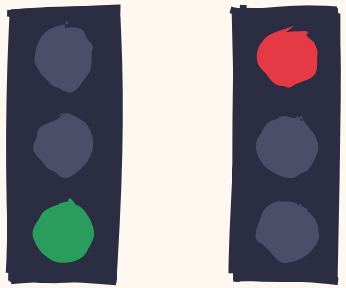
A junction has **two** lights, one per road. The safe sequence:

1. Road A **green**, Road B **red**
2. Road A **amber** (3 s), Road B still red
3. **Both red** for 1 s (clears the junction)
4. Road B green, Road A red … and repeat.

No matter what the sensors say, the code must **never** let both be green.

Interlock
Engineers call a rule that must ALWAYS be true a safety interlock.

Example: taking turns on a slide



A green

B red

At the playground slide, one child goes, then the next. If two go at once, they crash. The 'both red' moment is like waiting until the first child is off the bottom.

Objective 2 · Count cars

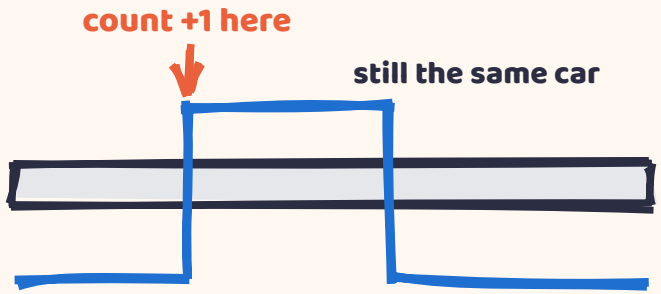
Count arrivals, not 'car is there'

The idea

The IR sensor says **LOW** when something is in front of it. If we add 1 **every time** we see LOW, one car parked there would count 1000s!

So we remember the **last** reading and count only when it changes from **no car → car**. This is called **edge detection**.

Example: a turnstile



A turnstile at a stadium clicks once per person, however long they stand in it. It counts the moment someone pushes through, not the time they spend there.

LEARN

Objective 3 · Smart decision

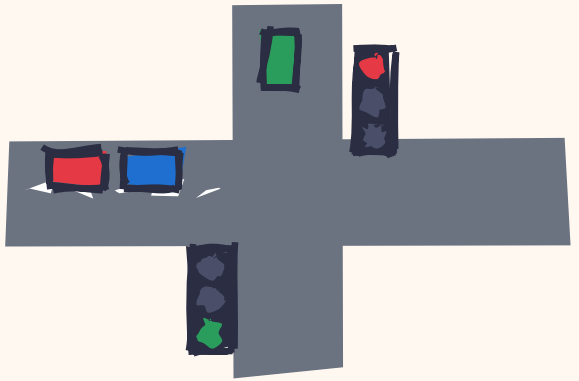
Smart and fair

The idea

Smart rule: when it's time to choose, give green to the road with **more waiting cars**.
Fair rule: a road may wait **at most 20 seconds**; then it gets green even with fewer cars.
Safety rule: always amber → all red → green.

When a road gets green, its waiting count goes back to 0 (those cars have gone).

Example: a school canteen



A: 6 cars

B: 1 car

The canteen opens the counter with the longest queue, but it would be unfair if one short queue never got served. So it also says: nobody waits more than 5 minutes.

Real world

Smart cities



Smart signals
Cameras and road loops count cars at busy junctions



Emergency priority
Ambulances can turn lights green ahead of them



Pedestrian buttons
Crossing time can extend for slower walkers



Cleaner air
Less waiting means less fuel burnt and less smoke

? Islamabad and Lahore have busy junctions. Where near your school would a smart light help most?

Safe City
Cities like Islamabad and Lahore use traffic cameras in 'Safe City' projects to watch and manage traffic.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Wire two traffic lights
2. A safe timed sequence
3. Human junction
4. Our junction kit (per group)
5. Count the cars
6. Make it smart
7. Traffic tests

MAKE

Objective 1 · Safe sequence · Build step 1 of 4

Wire two traffic lights

Part	Arduino
Road A: red, amber, green	pins 2, 3, 4
Road B: red, amber, green	pins 5, 6, 7
Each LED	+ 220 Ω to GND
IR sensor A / B OUT	pins 8 / 9
OLED SDA / SCL	A4 / A5
Buzzer	pin 11



Tidy wiring

Use red, yellow and green wires for the matching LEDs. Debugging is much easier!

Objective 1 · Safe sequence · Build step 2 of 4

A safe timed sequence

Loop

- A green ON, B red ON (all others OFF)
- wait 5 s
- A amber ON, B red ON
- wait 3 s
- A red ON, B red ON (all red)
- wait 1 s
- A red ON, B green ON
- wait 5 s
- A red ON, B amber ON
- wait 3 s
- all red, wait 1 s

Check every step

At each step, can you find a moment where BOTH roads show green? If yes, it's a crash!

Make a function

Make 'setLights(A, B)' so each step is one block. Fewer chances for mistakes!

Class game

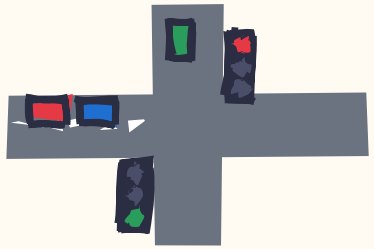
Human junction

How to play

1. Mark a crossroads on the floor with tape. Two students are the **traffic lights** with red/amber/green cards.
2. Other students are **cars**, queuing on both roads. Send more cars down one road.
3. Round 1: lights change every 10 s (timer).
4. Round 2: a 'sensor' student counts the queues and tells the lights which road is busier.
5. Which round moved more cars?

Then discuss

In round 2, did one road ever wait too long? How could we make it fair?



Big idea

Smart lights move more cars, but they need a fairness rule too.

Build it

Our junction kit (per group)



Arduino Uno
The junction controller



6 LEDs + 220 Ω
Two traffic lights (R, A, G)



2 IR sensors
Count cars on each road



OLED display
Shows car counts



Cardboard
Roads, poles and junction



Buzzer
Crossing beeper

Bring small toy cars, or make some from card. The IR sensor sees anything that passes close in front of it.

Plan first
Draw your junction on paper: where do the lights and sensors go?

Objective 2 · Count cars · Build step 3 of 4

Count the cars

```
Setup: carsA = 0, lastA = HIGH
Loop
  set nowA to digital read pin 8
  if nowA = LOW and lastA = HIGH then
    change carsA by 1
  set lastA to nowA
(same for road B on pin 9)
  OLED clear · print 'A: ' carsA · print 'B: ' carsB · show
```

Test	Count shown
Push 5 cars past A	
Hold a car still 5 s	
Push 3 cars past B	

IR knob

Turn the sensor's little knob so it only sees cars on ITS road, not the other one.

Objective 3 · Smart decision · Build step 4 of 4

Make it smart

```
Loop
  (count cars as before)
  if green is A and ( carsB > carsA or waitB > 20 s ) then
    change to B (amber 3 s → all red 1 s → B green)
    set carsB to 0 · set waitB to 0
  if green is B and ( carsA > carsB or waitA > 20 s ) then
    change to A (amber 3 s → all red 1 s → A green)
    set carsA to 0 · set waitA to 0
```

Minimum green

Also keep green for at least 5 s, or the lights could flicker between roads!

Test like an engineer

Traffic tests

Situation	What happened	Pass?
Equal traffic both roads		
Lots of cars on A, none on B		
1 car on B, stream on A		B waits ≤ 20 s?
Hold a hand in front of a sensor		
Both green ever? (must be NO)		

Debug checklist

Count stuck? Check edge detection: is lastA updated every loop?
OLED blank? Check A4/A5 and address 0x3C.

Test the dangerous cases on purpose. Safety engineers try to break their own systems.

Objective 3 · Smart decision · extension

Add a pedestrian crossing

The idea

Add a button for walkers. When pressed: finish the current green, go all red, then beep the buzzer for 8 seconds while people cross.

Accessibility

Real crossings beep so blind people know when it's safe. Some also vibrate a button for people who can't hear.

Think

Should a pedestrian always get priority over cars? What if 50 cars are waiting?

Level up!

Traffic engineer challenges

Choose your challenge



Ambulance mode: a special button gives instant green (after a safe amber!) to one road.



Rush hour: after 5 minutes of busy traffic, make greens longer.



Statistics: show total cars per road and average wait on the OLED.



4-way: add a third road. How does the sequence change?

Think like an engineer

Why is the 'all red' moment important even though it seems to waste time?

Boss level

Make the junction count cars for a whole lesson and graph the results.

Think about it

Cameras at junctions

Helpful

Cameras can count cars, catch dangerous drivers and help ambulances.

But...

Cameras can also record everyone's car number and where they go. Who should see this data? How long should it be kept?

? Our IR sensor counts cars without knowing **whose** car it is. Is that a better design? Discuss in pairs for 2 minutes.

Objective check

Can I really do it? Prove it.

01	Show your sequence. Prove both roads are never green together.	○ ○ ○
02	Push 5 cars past a sensor. Why does a parked car count only once?	○ ○ ○
03	Explain your smart rule AND your fair rule.	○ ○ ○
04	Read out your test table. Which test was hardest to pass?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Traffic jam challenge: the teacher sends lots of cars down one road. Whose junction copes best?

Show what you know

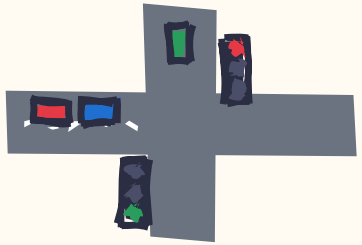
What did we learn?

- [x] Safety first: never both green; amber, then all red
- [x] Edge detection counts arrivals, not standing cars
- [x] Smart lights give green to the busier road
- [x] Fairness: no road waits forever
- [x] Test the dangerous cases on purpose

Exit ticket: why do we count when the sensor changes from HIGH to LOW, not whenever it is LOW?

At home

Watch a junction near home (from a safe place!). How long is each green? Is it timed or smart?



Next class:
Smart Kitchen Scale!